SELECT statement

The **SELECT statement** in SQL (Structured Query Language) is the most fundamental and commonly used command. Its primary purpose is to **retrieve data from one or more tables** within a database.

Core Function and Syntax

The **`SELECT`** statement allows you to specify which columns you want to see, which table(s) the data should come from, and which rows we want to filter for.

The basic syntax structure is: SQL

```
SELECT column1, column2,...
FROM table_name
WHERE condition;
```

Example:

```
SELECT CustomerID, CustomerName, Address, City,
PostalCode, Country FROM Customers WHERE Country=
'Mexico';
```

Result:

CustomerID	CustomerName	Address	City	PostalCode	Country
2	Ana Trujillo Emparedados y helados	Avda. de la Constitución 2222	México D.F.	05021	Mexico
3	Antonio Moreno Taquería	Mataderos 2312	México D.F.	05023	Mexico
13	Centro comercial Moctezuma	Sierras de Granada 9993	México D.F.	05022	Mexico
58	Pericles Comidas clásicas	Calle Dr. Jorge Cash 321	México D.F.	05033	Mexico
80	Tortuga Restaurante	Avda. Azteca 123	México D.F.	05033	Mexico

Key Clauses

The ***SELECT*** statement is usually composed of several clauses that define the query's criteria.

1. `SELECT` Clause

This is the required starting point.

- It specifies the **columns** we want to retrieve.
  - To select all columns from a table, we use the **asterisk (`*`)** wildcard: **`SELECT *`**.
  - We can also use **aggregate functions** (like `COUNT`, `SUM`, `AVG`, `MAX`, `MIN`) or **scalar functions** (like `UPPER`, `LOWER`) here.

Example:

```
SELECT CustomerID, CustomerName, Address, City,
PostalCode, Country FROM Customers;
```

2. `FROM` Clause

This is also required and follows the `SELECT` clause.

- It specifies the **table(s)** from which the data will be retrieved.
- It is also where we define **table joins** (e.g., `INNER JOIN`, `LEFT JOIN`) to combine data from multiple tables.

3. `WHERE` Clause (Optional)

This clause is used to **filter the records** (rows) based on a specified condition.

- Only rows that satisfy the condition will be included in the result set.
- It uses comparison operators (`=`, `>`, `<`), logical operators (`AND`, `OR`, `NOT`), and special operators (`LIKE`, `IN`, `BETWEEN`).

The SELECT DISTINCT statement in SQL is used to return only unique (non-duplicate) values in the specified column(s) within the result set.

Example: We want to retrieve all the different countries from the "Customers" table:

```
SELECT DISTINCT Country FROM Customers;
```

Result: Number of Records: 21

Country
Argentina
Austria
Belgium
Brazil
Canada
Denmark
Finland
France
Germany
Ireland
Italy
Mexico
Norway
Poland
Portugal
Spain
Sweden
Switzerland